# Flatten#

https://pytorch.org/docs/stable/generated/torch.nn.modules.flatten.Flatten.h

**Description:**

Flattens a contiguous range of dims into a tensor. For use with Sequential, see torch.flatten() for details. Input: (∗,Sstart,...,Si,...,Send,∗) (*, S_{\text{start}},..., S_{i}, ..., S_{\text{end}}, *)(∗,Sstart,...,Si,...,Send,∗),' where SiS_{i}Si is the size at dimension iii and ∗*∗ means any number of dimensions including none. Output: (∗,∏i=startendSi,∗)(*, \prod_{i=\text{start}}^{\text{end}} S_{i}, *)(∗,∏i=startendSi,∗). start_dim (int) – first dim to flatten (default = 1).

## Function Signature

```
class torch.nn.modules.flatten.Flatten(start_dim=1,
end_dim=-1)[source]#

Shape:

Parameters

Examples::

extra_repr()[source]#
```

**Returns:**

Tensor

## Code Examples

```
>>> input = torch.randn(32, 1, 5, 5)
>>> # With default parameters
>>> m = nn.Flatten()
>>> output = m(input)
>>> output.size()
torch.Size([32, 25])
>>> # With non-default parameters
>>> m = nn.Flatten(0, 2)
>>> output = m(input)
>>> output.size()
torch.Size([160, 5])
```

# Detailed Documentation

## Documentation

Flattens a contiguous range of dims into a tensor.

For use with Sequential, see torch.flatten() for details.

Input: (∗,Sstart,...,Si,...,Send,∗)(*, S_{\text{start}},..., S_{i}, ..., S_{\text{end}}, *)(∗,Sstart,...,Si,...,Send,∗),’ where SiS_{i}Si is the size at dimension iii and ∗*∗ means any number of dimensions including none.

Output: (∗,∏i=startendSi,∗)(*, \prod_{i=\text{start}}^{\text{end}} S_{i}, *)(∗,∏i=startendSi,∗).

start_dim (int) – first dim to flatten (default = 1).

end_dim (int) – last dim to flatten (default = -1).

Returns the extra representation of the module.

Runs the forward pass.